



Introduction

The Eclipse simulator suite consists of two separate simulators: Eclipse 100 specializing in black oil modeling, and Eclipse 300 specializing in compositional modeling.

Eclipse 100 is a fully-implicit, three phase, three dimensional, general purpose black oil simulator with gas condensate options. Eclipse 300 is a compositional simulator with cubic equation of state, pressure dependent K-value and black oil fluid treatments. Eclipse 300 can be run in fully implicit, IMPES and adaptive implicit (AIM) modes.

For large simulations the simulators can be run in parallel mode. The Parallel option is based on a distributed memory architecture implemented using MPI (message passing interface).

About this manual

This manual contains the technical descriptions of the principal features in both simulators. (The data file structure and the data requirements for each keyword are described separately in the *Eclipse Reference Manual*.) Some features are available in both Eclipse 100 and Eclipse 300, while others are available in only one of the simulators. A ‘flag table’ at the start of each chapter indicates which of the simulators contain the corresponding feature.

New and changed information in this release is shaded with a grey background.

For information on licensing see "[Licenses](#)".

In chapters that are relevant to both simulators, some paragraphs or sections may apply to only one of the simulators. These are distinguished by **margin notes** such as that containing the words Eclipse 100 as shown below.

Eclipse 100

Example of margin note.

Eclipse features

Free format input

Input data for Eclipse is prepared in free format using a keyword system. Any standard editor may be used to prepare the input file. Alternatively Petrel or Eclipse Office may be used to prepare data interactively through panels, and submit runs. Its help facility contains most of the Eclipse Reference Manual.

Phase and Fluid options

Eclipse 100

Eclipse 100 can be used to simulate one, two or three-phase systems. Two phase options (oil/water, oil/gas, gas/water) are solved as two component systems saving both computer storage and computer time. In addition to gas dissolving in oil (variable bubble point pressure or gas/oil ratio), Eclipse 100 may also be used to model oil vaporizing in gas (variable dew point pressure or oil/gas ratio).

Eclipse 300

Eclipse 300 can simulate using a cubic equation of state or pressure dependent K-values, as well as the black oil model. Four equations of state are available, implemented through Martin's generalized equation ([Martin, 1973](#)). The Redlich-Kwong, Soave-Redlich-Kwong, Peng-Robinson and Zudkevitch-Joffe cases of this equation may be used. A volume shift parameter may also be used with any equation. Different sets of equation of state parameters may be used for the reservoir and surface facilities. K-values may be supplied in table form or generated internally. See "[Equations of State](#)" for further information.

Eclipse 300

In [GASWAT](#) mode the solution of hydrocarbon components in the water phase is permitted. See "[GASWAT Option](#)" for further information.

Geometry options

Both corner-point and conventional block-center geometry options are available in Eclipse. Radial and Cartesian block-center options are available in 1, 2 or 3 dimensions. A 3D radial option completes the circle allowing flow to take place across the 0/360 degree interface (keyword [COORDSYS](#)).

The Eclipse corner-point geometry option allows extremely complex geometries to be constructed to give a faithful representation of the reservoir geology. Petrel can be used to prepare corner-point data for Eclipse. Petrel may also be used to display the grid in a variety of ways. For example, in a large 3D study, you may request a number of XZ cross-sections to be displayed simultaneously. Transparent areal overlays of the grid are useful for precise checking against geological maps.

Corner-point geometry is especially useful for highly faulted reservoirs. The grid may be distorted areally to fit along fault lines and displaced vertically to model complex scissor faults. The resulting non-neighbor grid transmissibilities are computed automatically by Eclipse. Flow across displacement faults is handled efficiently by the Eclipse solution procedures.

See "[Transmissibility Calculations](#)" for further information.

Fully implicit technology (Black Oil)

Eclipse uses the fully-implicit method to provide stability over long timesteps. Care is taken to ensure that the non-linear fully-implicit equations are solved precisely by reducing all residuals to very fine tolerances. Material balance errors (residual sums) are extremely small. Newton's method is used to solve the non-linear equations. The Jacobian matrix is fully expanded in all variables to ensure quadratic (fast) convergence. Various special methods are used to accelerate convergence in highly non-linear problems. The linear equations arising at each Newton iteration are solved simultaneously by Nested Factorization accelerated by Orthomin.

Traditionally, fully-implicit techniques are reserved for small coning studies, where many cell pore volumes of fluid may pass through small cells near the well bore during a single timestep. IMPES and semi-implicit techniques cannot be used on such problems unless the timesteps are reduced to impractically small values. When the fully-implicit method is used the resultant linear equations are strongly coupled and can only be solved efficiently by simultaneous methods and not by simple sequential methods. Small problems of this type can be solved efficiently by direct methods such as D4 Gaussian elimination. Iterative methods, required for large problems, such as SIP and LSOR, typically fail to converge. Thus most simulators cannot

apply fully implicit methods to large problems. In Eclipse, these restrictions are removed by Nested Factorization which solves large problems efficiently and reliably.

See ["Formulation of the Equations"](#) for further information.

Adaptive Implicit and IMPES (Compositional)

Eclipse 300

In a compositional model where the number of components and hence equations to be solved is greater than say 5 or 6, the cost of using a fully implicit technique can become prohibitive in both memory and CPU time. In Eclipse 300 this problem is tackled by using an adaptive implicit scheme, making cells implicit only where necessary.

In Eclipse 300 adaptive implicit, fully implicit or IMPES solution techniques may be selected. For larger compositional studies, it is likely that AIM will be the normal choice. The IMPSAT method may be used if large capillary pressures are present. Reservoir molar densities of each component are used as solution variables, along with the pressures. These yield a system that can be converted into a pressure equation and a set of conservation equations.

See ["Formulation of the Equations"](#) for further information.

Nested factorization

The linear equations arising at each Newton iteration are solved using Nested Factorization accelerated by Orthomin. This is a fast iterative technique devised for the solution of large sets of sparse linear equations. The technique conserves material precisely at each iteration. Material balance errors can therefore arise only as a result of non-linearities. Off band matrix elements arising from non-neighbor connections are included in the factorization procedure and are therefore handled with extreme efficiency. Computing costs increase only marginally faster than the number of active cells in the reservoir ($N^{5/4}$) in contrast with direct methods (N^3). Nested Factorization is therefore particularly well suited to the solution of large problems. Two and three phase problems are solved simultaneously. Further details of the method may be found in ["Solution of the Linear Equations"](#).

At each iteration, a new 'search direction' (approximate solution) is computed by Nested Factorization. The Orthomin procedure, which minimizes the sum of the squares of the residuals (errors), requires each new search direction to be orthogonalized to the previous search directions. Thus previous search directions must be stored, using up valuable computing memory. Fortunately it is usually sufficient to store only the last few search directions. The number of search directions is called `NSTACK` and you may set this in the `RUNSPEC` data section.

The solver is highly vectorized due to the use of active cell addressing. For example, all the Orthomin calculations on the stack are completely vectorized. Matrix multiplications are vectorized using the vector run technique. However, the tri-diagonal inversion at the heart of Nested Factorization, is inherently recursive and cannot be vectorized.

Non-neighbor connections

In conventional grid systems each cell has only one neighboring cell in each direction (two in each dimension). Thus in 2D grids, each cell can have up to four neighbors, and in 3D grids each cell can have up to six neighbors. Flow only takes place between neighboring cells.

In Eclipse, it is possible to connect selected pairs of non-neighboring cells, allowing fluid to flow directly from one cell of the pair to the other. Typical applications of non-neighbor connections are:

1. Completing the circle in 3D radial studies.

2. Displacement faults allowing flow between different layers across the fault.
3. Local grid refinement allowing fine definition near wells, including the ability to insert radial grids within a Cartesian system.

Eclipse generates the non-neighbor connections arising from the second application automatically, and calculates their transmissibilities. If you are using the Local Grid Refinement option (see "[Local grid refinement](#)"), Eclipse also handles the non-neighbor connections arising from the third application automatically. For a global radial grid, the [COORDSYS](#) keyword is used to determine whether the circle is completed, while for local radial grids, the circle is completed automatically.

Non-neighbor connections give rise to off-band elements in the Jacobian matrix and these are included in the procedures used to solve the linear equations.

Run-time dimensioning

All of the internal arrays in Eclipse are dimensioned at run time to minimize the use of computer memory. For example, each element of the Jacobian is a 2×2 matrix in two-phase runs and a 3×3 matrix in three phase runs. A two dimensional problem gives rise to a Jacobian matrix with 5 bands, while a 3 dimensional problem gives rise to a Jacobian matrix with 7 bands. Clearly it would be inefficient to dimension the internal arrays to cover the worst possible case. Most of the data required for Eclipse to compute the size of its internal arrays is provided in the [RUNSPEC](#) data section. Some additional information (relating to automatically generated non-neighbor connections, inactive cells and Vertical Equilibrium options for example) is deduced from data provided in the [GRID](#) section. Space is saved by not storing unnecessary data for inactive cells.

Vertical equilibrium

Eclipse 100

Eclipse 100 contains a unique Vertical Equilibrium option for modeling segregated flow in three dimensions. The option may be used with either block-center or corner-point geometry. With corner-point geometry, it takes account of the shape and orientation of each distorted grid block. A mixing parameter enables you to specify the degree of fluid segregation. At one extreme, total fluid segregation is assumed, while at the other extreme the fluids are assumed to be totally dispersed throughout each grid block. The VE facility takes account of hysteresis effects arising, for example, when oil invades the water zone and retreats, leaving a residual critical oil saturation.

It is also possible, using the Compressed Vertical Equilibrium option, to make the assumption that the entire reservoir section is in vertical equilibrium. This enables Eclipse 100 to collapse each column of grid cells into a single cell for calculation purposes. The variation in properties with depth is accounted for by using tables of properties for the compressed single cells.

See "[Vertical Equilibrium](#)" for further information.

Dual porosity

The Eclipse Dual Porosity/Permeability option is suitable for simulating highly fractured reservoirs. Each grid block is represented by a matrix cell and a fracture cell. In the Dual Porosity option, flow through the reservoir takes place through the fracture cells only. Flow also takes place between each matrix cell and its corresponding fracture cell. Flow does not take place between neighboring matrix cells. In the Dual Porosity/ Permeability option flow may also take place between neighboring matrix cells.

The Dual Porosity option can also model the effects of gravity imbibition/drainage, molecular diffusion between the matrix and fracture cells.

Eclipse 100

In addition Eclipse 100 can also model the viscous displacement of fluid to and from the matrix.

See ["Dual Porosity"](#) for further information.

PVT and rock data

Eclipse honors the pressure and saturation function data precisely as you specify it. It does not follow the practice, common in the industry, of smoothing the data by interpolating to a fixed number of equally spaced saturation intervals.

Different table numbers can be used for different parts of the reservoir. Thus, for example, separate saturation tables can be entered for each type of rock.

Directional relative permeabilities

Relative permeability curves may be specified separately for flow in the horizontal and vertical directions. You have the option of specifying 3 (X,Y,Z) or 6 (+X,-X,+Y,-Y,+Z,-Z) relative permeability curves for each saturation function region, rather than the usual single curve. This additional degree of freedom enables you to improve the physics of a simulation study in a relatively simple manner. For example, most reservoir cells have small vertical dimensions and large areal dimensions, and horizontal flow is more realistically modeled if the horizontal relative permeability is small below the Buckley-Leverett saturation. Directional relative permeabilities are also required for the Eclipse pseudo option that may be used on a three dimensional coarse grid.

Saturation table scaling

The connate, critical and maximum saturation end points of the relative permeability and capillary pressure curves may be input separately for each grid cell within the reservoir or, alternatively, as a series of depth tables for regions within the grid. The scaling option allows you to specify relative permeability and capillary pressure data which are functions of a normalized saturation. In addition, it allows the modeling of reservoirs where depth variations occur in the initial critical or connate fluid saturations. You may specify that the saturation table scaling is to be isotropic, distinguish between coordinate lines (X,Y,Z) or distinguish between coordinate directions (+X,-X,+Y,-Y,+Z,-Z).

The saturation table scaling facility may be used to equilibrate a reservoir model with the correct initial quantities of mobile fluid-in-place by means of a special correction applied at the fluid contacts.

See ["Saturation Table Scaling"](#) for further information.

Hysteresis

Both relative permeability and capillary pressure hysteresis effects can be modeled.

Eclipse 100

In Eclipse 100 there is a choice of two models for relative permeability hysteresis in the non-wetting phase: Carlson's model and Killough's model. There is also an option of using Killough's wetting phase hysteresis model.

Eclipse 300

In Eclipse 300 there is a choice of Killough's model and Jargon's model in the non-wetting phase.

Eclipse 100

An alternative three phase hysteresis model is available, which is applicable in cases where all three phases are present together, for example in a water alternating gas (WAG) flood (see ["Hysteresis in WAG Floods"](#)).

See ["Hysteresis"](#) for further information.

Rock compaction

The Rock Compaction option models the collapse of pore channels as the fluid pressure decreases. The process may be reversible, irreversible or hysteretic as required. The facility is particularly useful in dual porosity runs.

Eclipse 100

In addition in Eclipse 100 the Water Induced Compaction facility enables the mechanical effect of water on some rock types (for example chalk) to be taken into account.

See "[Rock Compaction](#)" for further information.

Tracer tracking

The Tracer Tracking option is a general facility to determine the movement of ‘marked’ fluid elements during a simulation run. It may be used, for example, to differentiate the movement of water injected by different wells and initial aquifer water, or to predict the variation in salinity or concentrations of other chemical species.

Eclipse 100

Tracers may also be defined to exist in a hydrocarbon phase, and used to model the exchange of the fluid elements between the free and solution states whenever mass transfer takes place in the associated hydrocarbon phase. The Tracer option is therefore useful in predicting the movement of the initial free and/or solution gas. Initial sulfur content may be defined as a function of depth and tracked with the movement of oil.

Eclipse 100

An option to control the effects of numerical diffusion on tracer interfaces is available.

See "[Tracer Tracking](#)" for further information

API tracking

Eclipse 100

Unlike the ‘passive’ tracer tracking options described above, API Tracking is ‘active’. It takes full account of the mixing of oils with totally different PVT properties. This contrasts with the conventional practice of assuming that the oil magically assumes the formation volume factor, viscosity, density and bubble point of the cell into which it flows. Initial API may be defined as a function of depth in each equilibration region or independently for each cell. A fourth equation is solved at each timestep to update APIs.

See "[API Tracking](#)" for further information.

Brine tracking

The brine tracking facility is an ‘active’ tracer in the water phase. The water properties are tabulated as a function of the salt concentration. Hence if injected water has a different salinity from the original reservoir water, the different water density and viscosity can be accounted for.

See "[Brine Tracking](#)" for further information.

Temperature model

Eclipse 100

The Temperature model is designed to model the cooling of the reservoir when cold water is injected into a relatively hot reservoir. An energy balance equation is solved after the main flow equations at each timestep, to determine the temperature in each grid block. Both the oil and water viscosities are modified as a function of temperature for the subsequent timestep.

See the "[Temperature option](#)" for further information.

Eclipse 300

In a compositional simulation the temperature option invokes the thermal option, see "[Thermal option](#)", this can result in a number of important differences, please see the section "[Differences between the temperature option in Eclipse 100 and Eclipse 300](#)" for more information.

Miscibility

Eclipse 100

Eclipse 100 has a three-component Miscible Flood model. This is intended for the simulation of miscible gas injection processes. The model assumes that the reservoir fluids consist of three components: reservoir oil (stock tank oil + solution gas), injection gas (solvent) and water. The reservoir oil and solvent gas components are assumed to be miscible in all proportions. Physical dispersion of the miscible components is treated using the Todd- Longstaff technique by means of a mixing parameter which is specified on a region basis. The solvent gas gravity can differ from that of the solution gas, and the model also allows the oil/solvent screening effects at high water saturations to be studied. A special numerical diffusion control facility is available with the miscible option, and is particularly useful in studies using large values of the mixing parameter. See "[Miscible Flood Modeling](#)" for further information.

Eclipse 300

In a compositional simulation miscibility is predicted naturally as the surface tension between the two hydrocarbon phases drops to zero. However the simulator still needs to model the changing relative permeability of the miscible fluid. Eclipse 300 contains a comprehensive miscibility model. See "[Surface Tension Effects](#)" for further information.

Individual well controls

Eclipse has a comprehensive set of options for controlling individual wells. Producers can operate at a specified value of the oil rate, water rate, liquid rate, gas rate, linearly combined rate, reservoir fluid voidage rate, bottom hole pressure and tubing head pressure. The engineer supplies a target value for one of these quantities, and limiting values for any of the remaining quantities. The well operates at its specified target as long as none of the limits are violated. If a limit is going to be violated, the well automatically changes its mode of control to keep operating within its allowed limits. Efficiency factors can be applied to take account of regular downtime. The flow rates and pressures are calculated for the well under normal flowing conditions, but the cumulative flows are reduced according to the efficiency factor. Injection wells have a similar set of controls. Target or limiting values can be supplied for the injection rate at surface conditions, the injection rate at reservoir conditions, the bottom hole pressure and the tubing head pressure.

To aid the process of history matching, production wells can be placed under an additional type of control. Their observed oil, water and gas rates are input, and the well can be made to produce at the equivalent liquid or reservoir fluid volume rate. Thus the rate of pressure decline should be approximately correct even when the water cut and gas-oil ratio are not fully matched. Both the observed and calculated flow rates can be written to the SUMMARY file for graphical comparison.

Production wells can also be subject to an additional class of 'economic' constraints. A producer is automatically shut-in or re-completed if its oil, gas or liquid rate falls below an economic limit. If its water cut, gas-oil ratio or water-gas ratio exceeds the specified upper limit, the well is worked over or shut-in. The rate of doing workovers can be limited by rig availability. Water cut, gas-oil ratio and water-gas ratio limits, and minimum economic rate limits, can also be applied to individual connections, which is closed if their limits are violated. In addition there is an option to limit the water cut, gas-oil ratio or water-gas ratio from a well by successively reducing its production rate each time the limit is violated. In Eclipse 100 wells subject to a minimum tubing head pressure limit can have their productivity boosted by automatically switching on artificial lift when their oil or liquid flow rates fall below a specified limit, subject to constraints on the group or field capacity for artificial lift. Individual wells can be 'flagged' to make the simulation run stop if they are shut-in for any reason, or another well can be opened automatically. The run can also be made to stop if all the producers in the field, or in one or more nominated groups, are shut-in. Wells that are shut-in can be tested periodically to see if they can flow economically again.

See ["Well controls and limits"](#) for further information.

Group and field production controls

The simulator contains facilities to control the collective behavior of groups of wells, or the field as a whole. The overall production rate of oil, water, gas, liquid or reservoir fluid (that is voidage) from one or more groups can be made to meet a specified target by one of two methods: prioritization or guide rate control.

With prioritization, wells are ranked in order of priority, and are turned on in decreasing order of priority until a group rate limit is exceeded. The well that exceeds the group rate limit has its rate cut back to meet the limit. The remaining lower-priority wells in the group are kept closed until they are needed. Well priorities can either be set by you, or calculated at specified intervals from a general formula with user-defined coefficients.

With guide rate control, the group's target rate is apportioned between all its producers in proportion to either their production potentials or specified 'guide rates', but ensuring that no well violates its individual flow rate or pressure limits. When the group no longer has enough potential to meet its production target, the production rate will decline. However, the simulator can postpone the decline by automatically opening new wells from a drilling queue whenever extra potential is required, subject to constraints on the drilling rate and the maximum number of open wells allowed in each group. In Eclipse 100 additional actions for maintaining the production rate (reperforating, retubing, switching on lift and changing to a lower THP limit) can be performed through 'group production rules'.

Upper limits can also be applied to groups' production rates of oil, water, gas, liquid and reservoir fluid. There is a choice of actions that are taken whenever one of these limits is violated, ranging from a work-over of the worst-offending well to the application of group control to hold the group's production rate at the limiting value. In addition, a set of 'economic constraints' equivalent to those described above for individual wells can be applied to the overall production behavior of groups.

The same set of controls and limits can also be applied to the overall field production. Under guide rate control, a target production rate for the field can either be apportioned between the groups in proportion to specified group guide rates, or apportioned directly between the wells in proportion to their production potentials or guide rates. The field production rate can be controlled simultaneously with the production from one or more of its constituent groups. The identity of the controlled phase (oil, water, gas, liquid or reservoir fluid) need not be the same in each case. For example, the field can produce oil at a required rate while one of the groups is producing at its maximum gas handling capacity. Thus the field produces at its target rate while each group honors its own set of constraints, and while each well honors its individual pressure and rate constraints.

See ["Group and field control facilities"](#) for further information.

Multi-level grouping hierarchy

Groups can be arranged into a multi-level tree structure of any required depth, by declaring some groups to be 'children' of higher level groups. All group controls and limits can be applied at any level (including the field, which can be regarded as a group at level zero at the top of the tree).

Group injection controls

Groups, and the field, can be given targets and limits to control their injection rates. The engineer can set target or limiting values for any of the following quantities:

- Surface injection rate of a particular phase

- Total reservoir volume injection rate of all phases.
- Re-injection fraction of a particular phase.
- Total voidage replacement fraction by all injected phases.

Thus for example a group can be made to re-inject a set proportion of its produced gas, and to inject water to make up an overall reservoir volume injection target or voidage replacement fraction.

A group injection target is apportioned between the wells in proportion to either their injection potentials or specified guide rates, subject to the wells' individual flow and pressure limits. A target for a high level group (or the field) can optionally be apportioned among its subordinate groups in proportion to their group guide rates, which can be set directly or calculated to equal their production voidage rates or voidage replacement needs. If a group (or the field) has insufficient injection potential to meet its target, the drilling queue is scanned for a suitable injector to open.

Eclipse 300

In compositional cases an injected gas stream may be specified explicitly, or be defined by the production from a well or group. It may also be a mixture of several such sources. The injection rate may be limited by the availability of gas from the specified source, although advance gas and make-up gas streams can be defined to augment the supply.

See ["Group injection control options"](#) for further information.

Sales gas production control

The Sales Gas Production Control facility provides a means of controlling the export of associated gas from an oil field, which may at the same time have controls on the oil production rate. The sales gas production rate of a group or the field is defined as its total gas production rate, minus its total gas injection rate, minus the gas consumption rate of the group and any subordinate groups, plus the rate of gas import (if any) to the group or its subordinates. Individual gas consumption and import rates can be set for each group.

The sales gas production rate is controlled by re-injecting the surplus gas not required for sale. Sales gas control can be applied independently of any other production controls on the group or field, provided that there is enough injection capacity to inject the surplus gas. The group or field is automatically placed under gas re-injection control, and its target re-injection fraction is determined dynamically at each timestep to inject the surplus gas.

See ["Sales gas production control"](#) for further information.

Crossflow and co-mingling in wells

When a well is completed in more than one grid block, the flow rate from each grid block is proportional to the product of three quantities:

1. The transmissibility between the grid block and the well bore.
2. The mobility of the phase in the grid block at the perforations.
3. The pressure drawdown between the grid block and the well bore.

Eclipse takes full account of all three quantities when apportioning a well's target flow rate between its various grid block connections. This is especially important in cases where a well is completed in two or more layers of a reservoir with poor vertical communication, where the drawdown at each layer can be substantially different.

In certain situations the drawdown in one layer may have the opposite sign to the drawdown in the other completed layers. The simulator will then allow crossflow to take place between the reservoir layers

through the well bore. A material balance constraint is observed within the well bore at formation level, so that the mixture of phases flowing out of the well bore corresponds to the average phase mixture entering the well bore. Eclipse can handle crossflow in any type of well operating under any mode of control, including the case when the well is plugged off above the formation while the completions remain open to the reservoir.

The crossflow facility can be turned off if the engineer does not wish this to occur.

See ["Well Inflow Performance"](#) for further information.

Highly deviated and horizontal wells

There is no restriction on the location of the completions within each well. A highly deviated well can be completed in a number of grid blocks that do not all belong to the same vertical column. In fact a well can be completed in several grid blocks within the same layer, enabling the simulator to model wells completed in horizontally-adjacent grid blocks. This feature is also useful in 3D radial coning studies where the well connects with all the innermost, wedge-shaped, blocks in any given layer. Multisegment Wells (see ["Multisegment wells"](#)) may be used to investigate the effects of friction pressure losses along the well's perforated length, which could be important for horizontal wells.

Special facilities for gas wells

Eclipse has two facilities for improving the accuracy of modeling gas inflow. Firstly, the non-Darcy component of the pressure drawdown resulting from turbulence near the well bore is modeled by a flow-dependent contribution to the skin factor. The simulator calculates the turbulence skin factor and the gas inflow rate in each completion by a method that ensures that they are fully consistent with each other. The calculation is performed fully implicitly, to maintain stability over long timesteps. These precautions are especially important when the pressure drop due to turbulence is a significant proportion of the total drawdown.

Eclipse 100

The second facility is the option to use a special inflow equation. Eclipse 100 offers the choice of using the Russell-Goodrich equation or the gas pseudo-pressure equation in place of the conventional inflow performance relationship for free gas flow. Both equations make allowance for the variation in the viscosity and the density of the gas over the pressure drop between the grid block and the well bore. For the Russell-Goodrich equation, these quantities are determined at the average pressure between the grid block and the well bore, instead of at the pressure of the grid block. For the pseudo-pressure equation, these quantities are replaced by an average value integrated between the grid block and well bore pressures.

In gas condensate simulations the Generalized Pseudo-Pressure method of Whitson and Fevang can be employed. Here, the total hydrocarbon mobility is integrated between the grid block pressure and the well bore pressure, taking account of changes in relative permeability caused by condensate dropout.

See ["Special inflow equations"](#) for further information.

Fine grid equilibration

The initial pressures and saturations in the reservoir grid blocks can be calculated by the 'equilibration' facility. Each phase in the reservoir is given a hydrostatic pressure gradient, dependent on its local density. The saturations are determined from the local pressure differences between the phases, using the capillary pressure tables.

A common practice is to set the fluid saturations within each grid block equal to the local saturations at the center of the block. However, this procedure can produce significant errors in the Fluids-in-Place estimation when a large grid block contains a fluid contact or part of a transition zone. Eclipse has an

option to improve the accuracy of the Fluids-in-Place calculation in this situation. The fluid saturations are determined at a number of levels within the block, and these are averaged to obtain the overall fluid saturations within the block. The method can also take account of the shape of sloping grid blocks, by weighting the fluid saturations at each level within the block with the area of the horizontal cross-section of the block at that level.

See ["Initializing the Study"](#) for further information.

Aquifers

Aquifers may be represented by a choice of two analytic models, a constant flux model, two constant head/pressure models and a numerical model. The analytic models are the Fetkovich aquifer and the Carter-Tracy aquifer. The constant flux model allows you to specify the flow per unit area at any time during the simulation. The constant head/pressure model allows you to specify the head or pressure at any time during the simulation. The numerical model consists of a sequence of aquifer cells connected together in one dimension. The innermost cell of the numerical aquifer sequence may be connected to any number of reservoir cells. You can specify attributes separately for each aquifer cell, for example depth, dimensions, porosity and permeability, giving you complete flexibility in defining each aquifer.

See ["Aquifer Modeling Facilities"](#) for further information.

Polymer flood model

The polymer flood option uses a fully implicit model to allow studying the detailed mechanisms involved in polymer displacement processes including sensitivity to salinity. The model accounts for the increase in viscosity of the aqueous phase due to the addition of the polymer solution. The Eclipse 100 model also accounts for the viscosity degradation due to non-Newtonian shear effects at high velocity.

The adsorption of polymer to the rock surfaces and the resulting reduction in aqueous phase relative permeability can be treated by the model. In Addition, in Eclipse 100 the effects of inaccessible pore volume on the flow of the polymerized solution are also included.

See ["Polymer flood model"](#) for further information.

Surface networks

The network option is designed to provide variable tubing head pressure limits to groups of wells, which vary with the groups' production rates according to a set of pipeline pressure loss relationships.

A simple application would be to model a subsea completion, where several wells produce into a single manifold which is connected to a separator by a flow line. The pressure in the manifold will depend on the total production rate, GOR and water cut of the wells.

For more complicated applications, a multilevel hierarchical network of pipelines can be built up. Each pipeline is given a vertical flow performance table (which can be produced by VFP *i*) to determine the pressure losses along it, in the same way that well bore pressure losses are calculated.

A network balancing calculation is performed at user-defined intervals to update the pressure at each node in the network. Separate networks can also be set up for water and gas injection, which can be useful for modeling pump or compressor performance into injection manifolds.

See ["Network option"](#) for further information.

Flux boundaries

The Flux boundary option allows runs to be performed on a small section of a field. A base run is performed for the full field study and time dependent flows across the boundary of the small section of the field are written to a flux file. Subsequent runs for the small section are performed, reading the flux file to establish boundary conditions for consistency with the full field model. This option is useful for history matching when only a few wells remain to be matched, and reduces the cost of repeated runs when only a small sector of the field remains to be resolved.

Eclipse 100

It may also be used in conjunction with the drainage region facility ([item 25](#) in the `OPTIONS` keyword) to reduce computing costs in sensitivity studies.

See "[Flux boundary conditions](#)" for further information.

Solvent model

Eclipse 100

The Solvent Model is a four-component extension of the Todd and Longstaff empirical miscible flood model. Eclipse solves 4 coupled equations fully implicitly for oil/water/gas/solvent. The oil/water/gas system in the absence of the solvent component reduces to the traditional black oil model. A solvent phase can be introduced into the model to miscibly displace the reservoir hydrocarbons. You can control the degree of miscibility using the Todd-Longstaff parameter, and the transition to miscibility with an input table of miscibility versus solvent saturation. Most potentially miscible fluids are only miscible at high pressure. This pressure effect can be modeled in Eclipse by defining a table of miscibility versus pressure.

See "[Solvent model](#)" for further information.

Gi pseudo-compositional model

Eclipse 100

The Gi pseudo-compositional model is a four-equation model which improves the modeling of volatile oil and gas condensate systems, without moving to a fully compositional description. The Gi parameter is a measure of the amount of dry gas that has contacted the original hydrocarbons. The saturated R_s and R_v values and the B_o and B_g of the oil and gas are supplied to Eclipse as functions of both pressure and Gi. For example the saturated R_v value of the gas will reduce with Gi; thus the gas is able to contain less vaporized oil. This effect overcomes one of the problems associated with black oil models, namely that of predicting overoptimistic recoveries in gas cycling studies. The data required by Eclipse can be generated from an equation-of-state description of the reservoir fluids using the PVT program.

See "[Gi pseudo-compositional model](#)" for further information.

Surfactant model

The surfactant model allows modeling of enhanced water floods where the added surfactant reduces the capillary pressure between the water and oil phases. The reduction in capillary pressure allows a greater proportion of the oil to be displaced, thus reducing the residual oil saturation. The Surfactant flood option models the reduction in the residual saturation and the associated modification of the relative permeability curves as a function of the capillary number (that is the ratio of viscous to capillary forces). The effect of the surfactant concentration on the water viscosity, and the adsorption of surfactant onto the rock, can be treated by the model.

See "[Surfactant model](#)" for further information.

Environmental tracer

Eclipse 100

The environmental tracer option is a set of extensions of the tracer tracking facility that can be used to model environmental problems, for example, the transport of a radioactive contaminant in ground water flow. At present the extra physical models are tracer adsorption and tracer decay. Up to fifty tracers can be tracked independently, each with its own adsorption and/or decay model.

See "[Environmental tracers](#)" for further information.

Coal bed methane

The Coal bed methane option allows the modeling of gas production (methane) from coal seams. Gas present in the coal matrix diffuses into a fracture system and can be produced using traditional oil field technology. The option models the diffusion process and the flow of water and gas within the fracture system. It is also possible to model enhanced recovery mechanisms using a third conservation equation to represent an injected gas, typically CO₂.

See "[Coal bed methane model](#)" for further information.

Foam model

The foam model can be used to screen fields for possible foam injection projects. It allows modeling the effect of foam on reducing gas mobility, this effect can be used to slow the breakthrough of injected gas or to reduce the production of the gas cap.

The Eclipse 100 foam model is a simple model of the foam injection process. The foam is modeled as a tracer in either the gas or the water phase, which reduces the gas mobility. The foam quality can degrade due to adsorption and decay, the decay rate being a function of both the water and oil saturations. In addition, the effectiveness of the gas mobility reduction is a function of pressure and the shear rate.

The Eclipse 300 foam model allows for greater flexibility and detail for modeling the effects of foam in the reservoir. Foam is modeled as a water component, which might be injected or generated in situ following surfactant flooding. Foam adsorption, decay and generation can be included in the model using chemical reactions.

See "[Foam model](#)" for further information.

Gas calorific value control

Eclipse 100

The gas calorific value control option controls the mean calorific value of gas produced from the field, while simultaneously controlling the gas production rate. The calorific value of the produced gas can either be specified independently for each well, or equated to the concentration of a gas phase tracer. A target mean calorific value can be set for groups that are operating at a rate target under guide rate group control (keyword [GCONPROD](#)). Eclipse attempts to keep the group operating at both targets simultaneously by dynamically adjusting the well guide rates to give an appropriate mixture of high and low calorific value gas. The range of well calorific values must of course span the group's target value. An additional option is the ability to specify a target energy rate (that is flow rate multiplied by calorific value) for the group. The facility is compatible with the Gas Field Operations Model, in that a group or field target for either gas production rate or energy production rate can be set according to the DCQ and the swing or profile factors.

See "[Gas calorific value control](#)" for further information.

Eclipse options

Eclipse options consists of a series of special extensions to Eclipse that increase the modeling capability of the program. These are:

- Local grid refinement and coarsening,
- Wellbore friction option,
- Gas field operations model,
- Gas lift optimization,
- Parallel options,
- Reservoir coupling,
- Multisegment wells.
- Thermal option

The special extensions may be purchased separately as required.

Local grid refinement

The local grid refinement option allows enhanced grid definition near wells. The local models may be 2D radial, 3D radial or 3D Cartesian. Local models may have more layers than the global model. Transmissibilities between the local models and the global model are computed automatically.

Eclipse 100

Greater efficiency may be achieved by solving each local model individually. In general, local models have shorter timesteps than the global model so that detailed coning effects can be resolved near troublesome wells without holding up the progress of the global field simulation.

It is also possible to solve the combined local and global system fully implicitly as a coupled system (“LGR in-place”). In Eclipse 300 all LGRs are solved simultaneously with the global grid.

Grid coarsening can also be used to amalgamate cells in regions of the field where accuracy is not so important.

Eclipse 100

An auto-refinement facility may be used to refine an existing global model as the reverse process to grid coarsening.

See ["Local grid refinement and coarsening"](#) for further information.

Wellbore friction

Eclipse 100

The wellbore friction option models the effects of pressure loss due to friction in the well tubing along the perforated length and between the perforations and the bottom hole reference point of the well. The facility is primarily intended for use with horizontal wells, in which frictional pressure losses may be significant over the horizontal section of the wellbore. It is important to consider friction losses when deciding on the optimum length and diameter of a horizontal well. The friction calculation is handled fully implicitly, to retain numerical stability.

See ["Wellbore friction option"](#) for further information.

Gas field operations model

The gas field operations model is a collection of facilities designed to model gas field production constraints and operations. These comprise:

- Using a seasonal profile of monthly multipliers to the mean rate or DCQ (Daily Contracted Quantity) to set the production target,
- Adjusting the DCQ each contract year to obey a swing factor requirement,
- Control and output of sales gas production,
- Estimation of delivery capacity at each timestep, allowing for the increased pressure losses through the network,
- Automatic operation of compressors in the network, whenever they are needed to achieve the production target.

Eclipse 100

See ["Gas field operations model"](#) for further information.

Gas lift optimization

The gas lift optimization facility determines how much lift gas to allocate to each well in order to meet well, group or field production targets. If production targets cannot be met, it determines how to make the best use of the available lift gas resources by allocating lift gas preferentially to the wells that can make the best use of it. The facility can also be used with the Network option to optimize the lift gas distribution within a small network of a few wells, taking account of the extra pressure losses in the flowlines due to the increased flows.

See ["Gas lift optimization"](#) for further information.

Parallel options

The parallel options allow Eclipse to take advantage of computers with more than one processor. The facility provides an efficient parallel solution method, using a domain decomposition technique, which divides the simulation grid into domains which are solved in parallel.

The facility can run on both shared memory and distributed memory machines.

See ["Parallel option"](#) for further information.

Reservoir coupling

Eclipse 100

The reservoir coupling facility provides a means of coupling a number of separate eclipse 100 simulation models to allow for constraints on their overall production and injection rates, and optionally the sharing of a common surface network. An oil or gas field may comprise a number of isolated reservoir units that are coupled only by global production and injection constraints. These may have been studied separately (for example, for history matching) with their own simulation models. The reservoir coupling facility provides a convenient way of combining these models to obey global production and injection constraints, with minimal change to their input data. The simulation models are run as separate processes, under the control of a master process which dictates their production and injection constraints at each timestep. Communication between the processes is handled by the MPI message passing system, which allows the individual simulation models to be run in parallel on different UNIX computers.

See ["Reservoir coupling"](#) for further information.

Multisegment wells

The multisegment well model provides a detailed description of fluid flow in the well bore. The facility is specifically designed for horizontal and multilateral wells, and to model 'smart' wells containing various types of flow control device, although it can of course be used to provide a more detailed analysis of fluid flow in standard vertical and deviated wells. A down-hole separator model is also available. The detailed description of the fluid flowing conditions within the well is obtained by dividing the well bore (and any lateral branches) into a number of segments. Each segment has its own set of four independent variables to describe the local fluid pressure, the flow rate and the flowing fractions of water and gas. The pressure drop along the well bore is calculated from the local hydrostatic, friction and acceleration pressure gradients. The pressure drop can be calculated from a choice of two built-in multi-phase flow models: a homogeneous flow model where all the phases flow with the same velocity, or a 'drift flux' model which allows slip between the phases. Alternatively, the pressure drop may optionally be derived from precalculated VFP tables, which can potentially offer greater accuracy and provide the ability to model chokes and other flow control devices. The Multisegment Well Model also enables complex crossflow regimes to be modeled more accurately, as the re-injected fluid mixture can vary with position along the well bore.

See ["Multisegment wells"](#) for further information.

Thermal option

Eclipse 300

The Thermal option turns Eclipse 300 into a thermal simulator that models the flow of steam as well as the flow of hot or cold water in an oil reservoir.

The thermal option runs in fully implicit mode. The production system model is the multilevel hierarchy in Eclipse 300, with an implicit crossflow well model. The thermal model will run in either a dead-oil or a live-oil mode. In the dead-oil model, hydrocarbon components are only present in the oil phase, and the gas phase consists only of steam. The live-oil model allows hydrocarbon components in both the oil and the gas phase, and water components in both the water and the gas phase.

See ["Thermal option"](#) for further information.

Ancillary programs

Ancillary programs available with Eclipse are described below.

VFPi

VFP *i* is an interactive program for preparing, examining and editing VFP tables. It calculates pressure drops in wells and pipelines, and generates vertical flow performance tables for input into Eclipse. The program contains extensive graphical facilities for examining tubing data and fluid properties, and plotting the results of individual pressure traverses and calculated vertical flow performance curves. A well IPR can be superimposed on the curves to provide an estimate of the flow rate at a given set of conditions. VFP tables can be read in and examined in cross section or 3D. There are facilities for transforming the tables in a number of ways, and visually editing individual points within a table. Additional features of VFP *i* include an enthalpy equation to calculate the temperature variation of the fluid along the tubing, and a facility to match calculated pressures with measured data by tuning a family of 'fitting factors'. VFP *i* can also perform pressure drop calculations for a compositional fluid, and produce compositional vertical flow performance tables for use with Eclipse 300.

VFP *i* offers a choice of seven multi-phase flow correlations to calculate the pressure traverse:

1. Aziz, Govier and Fogarasi,

2. Orkiszewski,
3. Hagedorn and Brown,
4. Beggs and Brill,
5. Mukherjee and Brill,
6. Gray,
7. Petalas and Aziz.

Different correlations may be used for different sections of tubing, enabling VFP i to handle horizontal and undulating pipelines as well as vertical well bores and risers. The calculations can include the effects of gas lift, a down-hole pump, a gas compressor, and a choke with both critical and sub-critical two-phase flow.

Additional programs

Programs, which may be purchased as special additions to the basic Eclipse system, are described below.

OpenEclipse

OpenEclipse enables Eclipse 100 and Eclipse 300 to be controlled by, and communicate interactively with, other applications. Although Eclipse is primarily used in a batch stand-alone mode, controlled solely by the contents of the input data file, there are many situations in which it would be advantageous to have Eclipse controlled by another software application. Examples include the requirement to have Eclipse tightly coupled to a surface gathering system model or a specialized production optimization application. It could also be useful to couple Eclipse to an interactive controller, to allow you to view the current status of the simulation and make well management decisions during the course of the run.

OpenEclipse was developed to address these requirements. It consists of subroutines in the code that communicate with an external application. Activation of OpenEclipse puts Eclipse into ‘listening mode’, awaiting commands from a suitably configured controlling program. The controlling program may set well and group operating constraints, interrogate well and group flows, control the advancement of the simulation run and control the output of reports and RESTART files. Communication between Eclipse and the controlling program is handled by MPI, which is provided with Eclipse.

Those wishing to develop their own applications to control Eclipse may obtain the OpenEclipse Developer’s Kit. This consists of full documentation of the communications dialog, a copy of the message passing systems annex with which to link their application, and a simple demonstration controller for testing the OpenEclipse interface.

FloGrid

FloGrid is an interactive 3D product that constructs fluid flow simulation grids and properties.

FloGrid imports geological data in most popular map and 3D geological model formats, including the POSC Rescue format. Simulation grids can be exported in Eclipse and other formats. FloGrid provides full 3D visualization of the data - wells, maps, geocellular models, faults and simulation grids.

FloGrid supports the generation of both structured and unstructured grids. Structured (Cartesian) grids are optimized to fit to selected faults and the boundary, with remaining grid nodes placed to minimize orthogonality errors. Faults that are not explicitly gridded may be automatically zig-zagged. For 3D geological models, simulation layering can, optionally, be determined using algebraic

and flow based techniques to automatically identify simulation layers that best capture the flow characteristics of the fine scale models.

Unstructured PEBI and tetrahedral (PetraGrid) grids can be tailored to honor all boundaries, wells and faults. They support rectangular and radial refinements around wells, and general refinements consisting of a rectangular, triangular or hexagonal grid within a specified polygonal region.

FloGrid also provides a suite of algebraic and flow based single phase upscaling tools that calculate appropriate simulation block properties from the fine scale geologic or stochastic property grids.

PVTi

PVT *i* is an interactive Equation of State (EoS) package used for the analysis of laboratory measurements performed to determine the phase behavior of reservoir fluids. The quality of the laboratory measurements can be tested through material balance checks. Laboratory experiments can be simulated using a variety of cubic EoS, and any discrepancies between calculated and measured data can be minimized by regression of one or more EoS parameters. The EoS model can then be used to generate data suitable for use in Eclipse 100 or 300 and VFP *i*.

SCAL

SCAL is a tool to help you effectively use laboratory derived relative permeability and capillary pressure measurements in reservoir simulation. The program has facilities to read in laboratory data, perform quality control such as curve smoothing, group data according to lithological parameters and end-point values, transform the laboratory data into rock curves suitable for Eclipse, and automatically assign these curves to grid cells on the basis of rules which you set up (for example, as a function of porosity / permeability / lithological parameters). The output consists of a series of [INCLUDE](#) files, for both the [PROPS](#) and [REGIONS](#) sections. The program has facilities for 3D visualization of simulation grids and three-phase relative permeabilities, and for experimenting with Eclipse end point scaling options. The behavior of SCAL can be extended or modified by the use of user-programmable command scripts to, for example, implement a company confidential algorithm.

Schedule

Schedule imports production data from a variety of common sources including PA, OilField Manager and Finder, and generates the corresponding Eclipse production control keywords. Production data can also be extracted from existing Eclipse models. The program has advanced graphic display features, which simplify the editing, validating and averaging of production data. Timesteps on which rates are averaged are based on a user-defined combination of calendar periods and reservoir events such as well completions, shut-ins and simulations.

All the main categories of production data necessary for simulation can be handled by Schedule. These typically take the form of well deviation surveys, historical production and injection volumes and completion data. A key feature of Schedule is the capability to generate accurate and representative [COMPDAT](#) keywords with time-varying connection factors calculated from perforation data specified in terms of measured depths and formations. Corrections are made for deviated wells, partial penetrations and multiple completions within a single cell.

Schedule has comprehensive facilities for creating prediction run controls for Eclipse. Controls can be set for wells, groups and the field. Wells can be created by defining IJK locations or by digitizing in the 3D Viewer, where simulation results can be used as a backdrop to aid placement.

Schedule can also prepare data for input to the Multisegment Well Model (see "[Multisegment wells](#)"). It can read data describing casing, liner and tubing characteristics and locations of chokes, packers and inflow control valves. This information can be used, with perforation data, to generate [WELSEGS](#) and [COMPSEGS](#) / [COMPSEGL](#) keywords describing the multisegment well.

FloViz

FloViz is an interactive 3D visualization system for the display and analysis of reservoir simulation results. It replaces RTView as the product of choice for reservoir visualization. FloViz has a simple to operate graphical user interface and provides faster 3D interaction and animation of simulation results. It runs on both PCs and workstations.

FloViz incorporates most of the RTView functionality with some key new features. The program allows multiple views (which can be tailored), through the use of slave viewers, of the reservoir with independent slicing, property, and time animation. Model rotation can be shared across the views. FloViz can display both structured and unstructured grids, and, also, streamlines. The program contains a host of new options to allow easier visual interpretation of the grid and its results.

Eclipse Office

Eclipse Office provides an interactive environment for the creation and modification of Black Oil and Compositional models, the submission and control of runs, the analysis of results and report generation. Data sets may be created using a PEBI gridding module, correlations for PVT and SCAL data, keyword panels, or input from other pre-processors. Panels exist for all Eclipse keywords. Tools are provided for the management of cases within a project. Run management is provided for submitting, monitoring and controlling simulator execution. New interactive graphics are used to view results.

SimOpt

SimOpt is aimed at reducing the man time taken during the history matching phase of a reservoir engineering study. It provides a means for obtaining sensitivities of Eclipse 100 and Eclipse 300 simulation results with respect to user-defined parameters, which may be specified on a regional basis. Parameters currently available include pore volumes, transmissibilities, fault transmissibilities, aquifer strength and dual porosity sigma factors. Sensitivities can be viewed directly, or used by a regression algorithm to improve the quality of a history match.

PlanOpt

PlanOpt is an interactive tool to assist you in choosing the potential locations of vertical production wells during development. PlanOpt applies pre-defined screening criteria to each column of grid blocks in a 3D simulation model to determine which could sensibly accommodate a production well. A vertical well is completed in each column of cells that satisfies the criteria. A simulation run is then used to rank the wells and the least effective wells are eliminated. This cycle is repeated until the optimum well locations have been identified.

FrontSim

FrontSim is a three-dimensional, two-phase fluid flow simulator based on a state-of-the-art streamline concept. FrontSim can perform simulations on large and complex reservoir models several orders of magnitude faster than standard finite difference simulators, and can achieve this without grid orientation effects or numerical dispersion.

The streamline concept is based on an IMPES (Implicit Pressure Explicit Saturation) solution. First the pressure is solved with an implicit numerical method and then the saturation equation is solved using an explicit method. The pressure is used to compute a velocity field which is, in turn, used to compute streamlines. The saturation equations are solved on the streamlines using a front tracking method.